Name: Aaron Ludwig A. Altar **ESP Activity#3**
Section: CPE4B **Date: Sept. 18, 2026**

3.1 LED Control Using BT Serial App

Code Snippet

```
// Activity #3
// 3.1 LED Control Using BT Serial App (ESP32)

#include "BluetoothSerial.h"

#if !defined(CONFIG_BT_ENABLED) || !defined(CONFIG_BLUEDROID_ENABLED)
#error Bluetooth is not enabled! Enable it in "Tools -> Partition Scheme"
#endif

BluetoothSerial SerialBT;

const int LED_PIN = 5;  // D5 - external LED w/ resistor
char incomingByte;

void setup() {
  Serial.begin(115200);
  SerialBT.begin("ESP32_LED_Control");  // Bluetooth device name shown in
app

  pinMode(LED_PIN, OUTPUT);
  digitalWrite(LED_PIN, LOW);

  Serial.println("Bluetooth started. Pair with 'ESP32_LED_Control'");
  Serial.println("Send '1' = LED ON, '0' = LED OFF");
}

void loop() {
  if (SerialBT.available() > 0) {
    incomingByte = SerialBT.read();

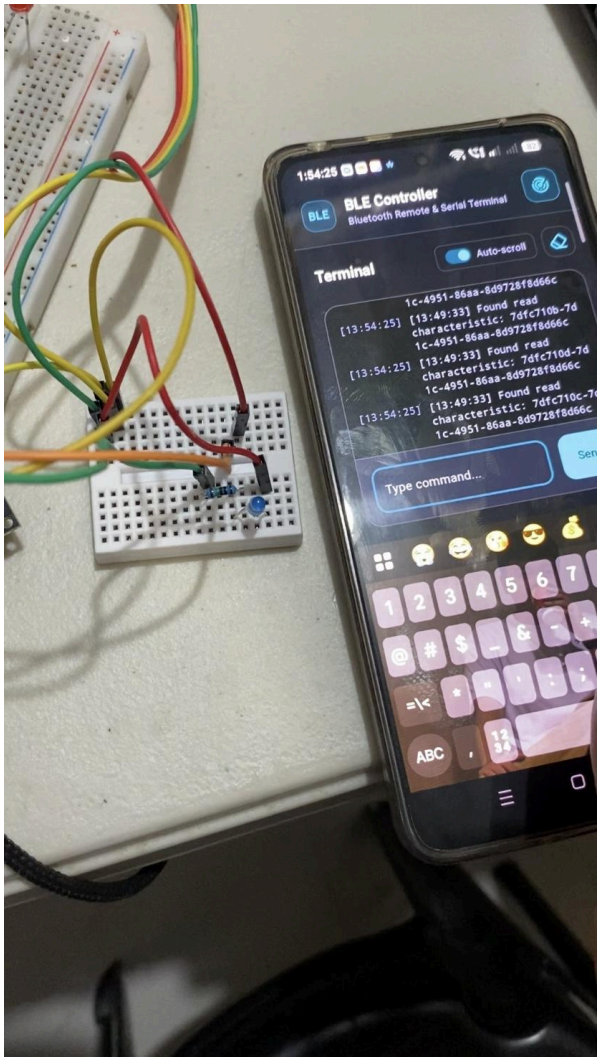
    if (incomingByte == '1') {
      digitalWrite(LED_PIN, HIGH);
      SerialBT.println("LED ON");
      Serial.println("LED ON");
    }
```

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```
    else if (incomingByte == '0') {
      digitalWrite(LED_PIN, LOW);
      SerialBT.println("LED OFF");
      Serial.println("LED OFF");
    }
  }
}
```

Actual Experiment





Video Link:

[3.1 LED Control using BT Serial App Video Link]

Name: Aaron Ludwig A. Altar **ESP Activity#3**

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3.2 Master Slave Communication Using BT

Code Snippet (Master)

```
// Activity #3
// 3.2 BT Master-Slave: MASTER (Central)

#include "BluetoothSerial.h"

BluetoothSerial SerialBT;

// Paste the slave's MAC address here (from the slave's Serial Monitor
output)
uint8_t slaveAddress[6] = {0xB4, 0xBF, 0xE9, 0x05, 0xED, 0x7A};

void setup() {
  Serial.begin(115200);
  SerialBT.begin("ESP32_MASTER", true);  // true = master mode

  Serial.println("Connecting to slave...");
  if (SerialBT.connect(slaveAddress)) {
    Serial.println("Connected! Type '1' or '0' and press Enter.");
  } else {
    Serial.println("Connection failed - check MAC address / slave is
powered on.");
  }
}

void loop() {
  if (Serial.available()) {
    char c = Serial.read();
    SerialBT.write(c);
  }
  if (SerialBT.available()) {
    Serial.write(SerialBT.read());
  }
}
```

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Code Snippet (Slave)

```
// Activity #3
// 3.2 BT Master-Slave: SLAVE (Peripheral)

#include "BluetoothSerial.h"
#include "esp_bt_device.h"

BluetoothSerial SerialBT;
const int LED_PIN = 5;  // D5 - external LED

void setup() {
  Serial.begin(115200);
  SerialBT.begin("ESP32_SLAVE");  // slave device name

  pinMode(LED_PIN, OUTPUT);
  digitalWrite(LED_PIN, LOW);

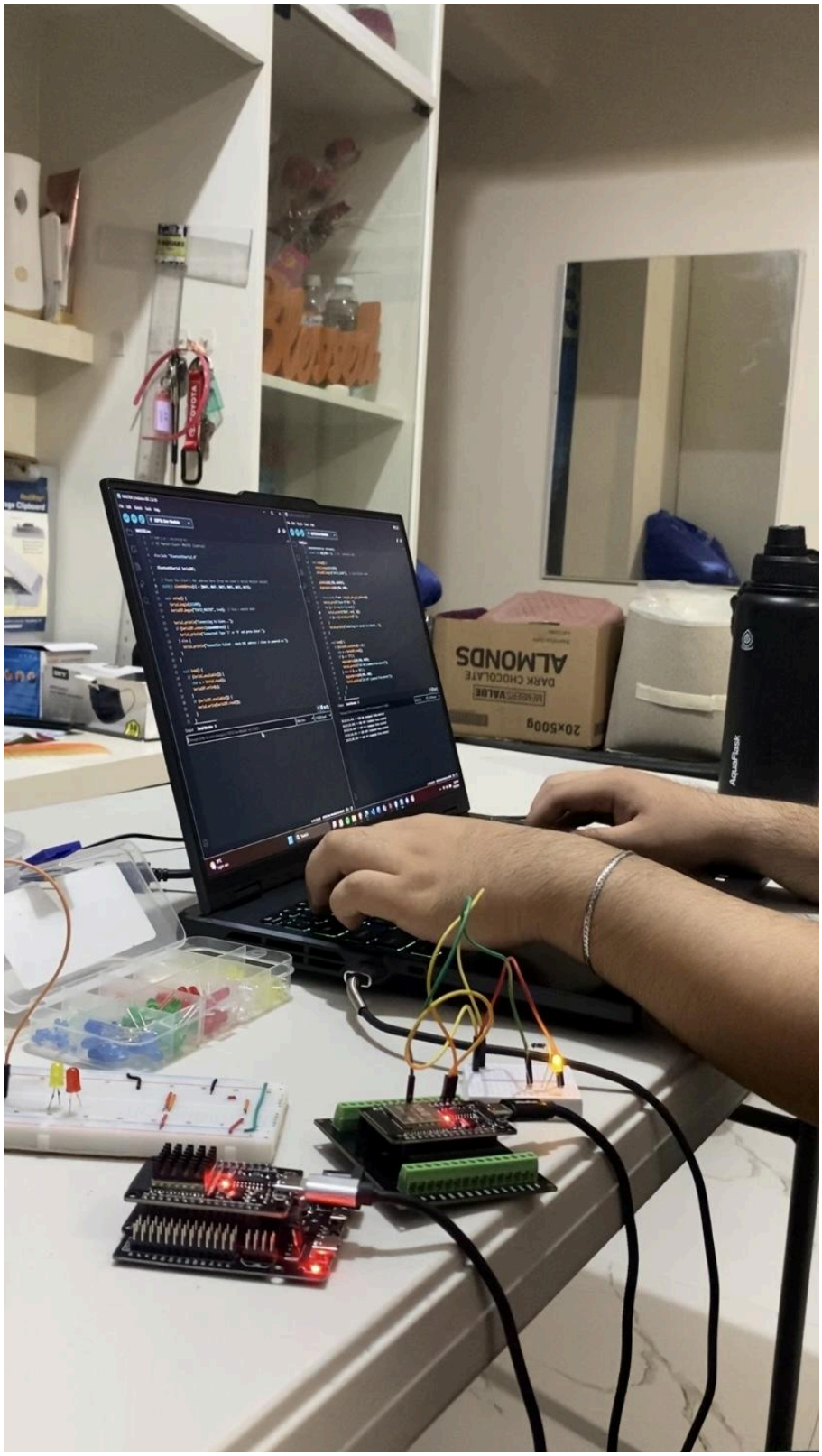
  const uint8_t* mac = esp_bt_dev_get_address();
  Serial.print("Slave BT MAC: ");
  for (int i = 0; i < 6; i++) {
    Serial.printf("%02X", mac[  i]);
    if (i < 5) Serial.print(":");
  }
  Serial.println("\nWaiting for master to connect...");
}

void loop() {
  if (SerialBT.available() > 0) {
    char c = SerialBT.read();
    if (c == '1') {
      digitalWrite(LED_PIN, HIGH);
      Serial.println("LED ON (command from master)");
    } else if (c == '0') {
      digitalWrite(LED_PIN, LOW);
      Serial.println("LED OFF (command from master)");
    }
  }
}
```

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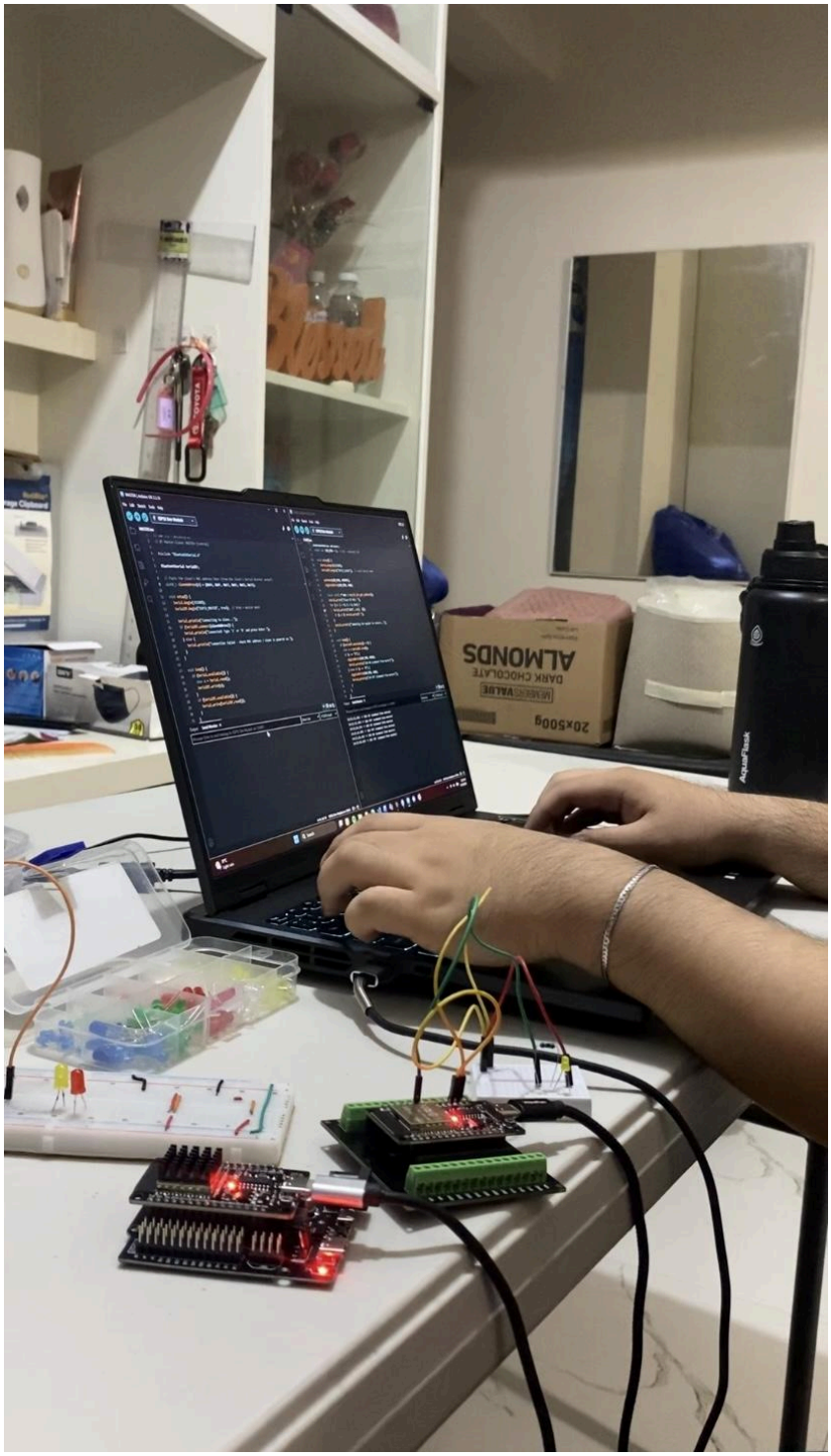
Actual Experiment (LED ON)



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Actual Experiment (LED OFF)



Video Link

[3.2 Master Slave Communication using BT Video Link]

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3.3 DHT11 Monitoring Using App Inventor/BT Serial App

Code Snippet

```
// Activity #3
// 3.3 DHT11 Monitoring Using BT Serial App (ESP32)

#include "BluetoothSerial.h"
#include <DHT.h>

#if !defined(CONFIG_BT_ENABLED) || !defined(CONFIG_BLUEDROID_ENABLED)
#error Bluetooth is not enabled! Enable it in "Tools -> Partition Scheme"
#endif

BluetoothSerial SerialBT;

#define DHTPIN 4        // D4 - DHT11 data pin
#define DHTTYPE DHT11

DHT dht(DHTPIN, DHTTYPE);

void setup() {
  Serial.begin(115200);
  SerialBT.begin("ESP32_DHT_Monitor");  // Bluetooth device name shown in
app
  dht.begin();

  Serial.println("Bluetooth started. Pair with 'ESP32_DHT_Monitor'");
}

void loop() {
  delay(2000);  // DHT11 needs ~2s between reads

  float humidity = dht.readHumidity();
  float temperature = dht.readTemperature();

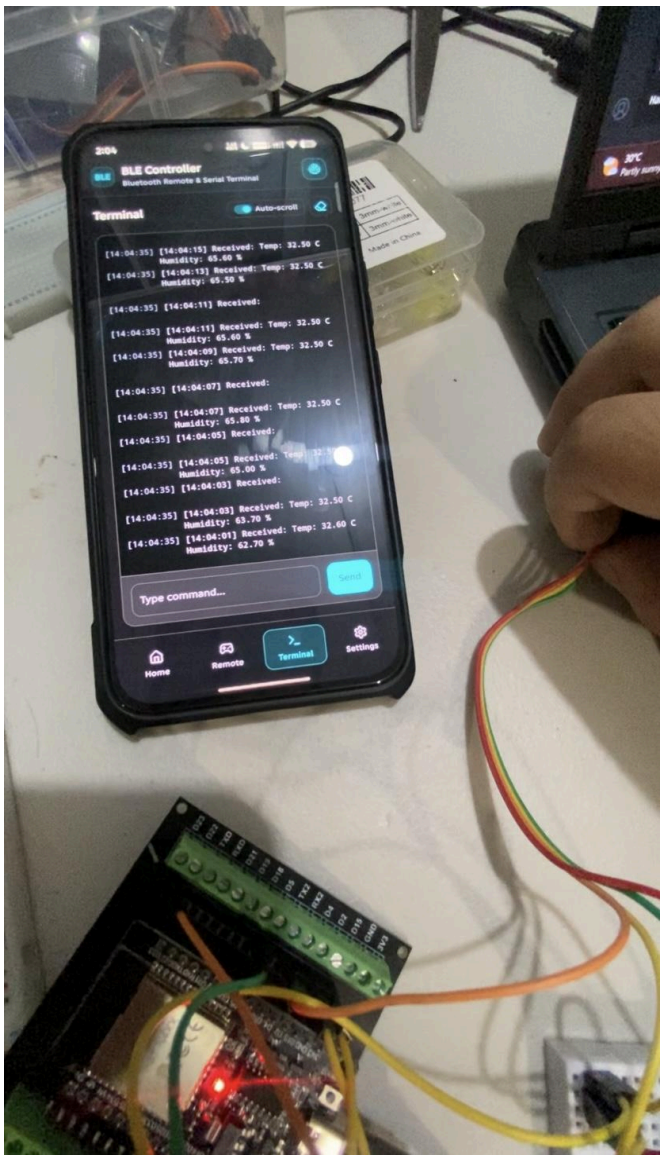
  if (isnan(humidity) || isnan(temperature)) {
    SerialBT.println("Failed to read from DHT sensor!");
    Serial.println("Failed to read from DHT sensor!");
    return;
  }
```

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```
  String reading = "Temp: " + String(temperature) + " C   Humidity: " +
String(humidity) + " %";
  SerialBT.println(reading);
  Serial.println(reading);
}
```

Actual Experiment



Video Link

[3.3 DHT11 Monitoring using BT Serial App Video Link]